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TASK 1: There is a need to overload = operator when it is being used to copy class objects which have dynamically allocated memory within them. Basically it is used when an already created object needs to be copied in to e.g:

```
Matrix m1(1,1);
Matrix m2(1,1);
m1 = m2; // = operator
```

Copy constructor is used during creation of objects. It is called during by value pass, return of object by value and when we create an object and initialize it at creation e.g:

```
Matrix m1;
Matrix m2 = m1; // Copy Constructor
```

Copy constructor is for initialization and = operator is for assignment.

TASK 2:

```
// Copy Constructor
Matrix(const Matrix &other)
{   this->rows = other.rows;
    this->cols = other.cols;
    this->allocateMatrix(this->rows, this->cols);
    for(int i=0; i<this->rows; i++)
    {  for (int j=0; j<this->cols; j++)
       {
            this->matrix[i][j] = other.matrix[i][j];
       }
}  }
```

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```
// = operator overload
Matrix ~~void~~ operator = (const Matrix &other)
{
    this -> deallocateMatrix();
    this -> rows;
    this -> cols;
    this -> allocateMatrix ( this -> rows , this -> cols);
    for(int i=0 ; i < this -> rows; i++)
    {
        for ( int j=0; j < this -> cols; j++)
        {
            this -> matrix [i][j] = other.matrix [i][j];
        }
    } return *this;
}
// * operator
Matrix operator *(const Matrix &other)
{
    if ( this -> cols != other .rows)
    {   cout << " Error \n";
        return *this;
    }
    Matrix temp (this -> rows , this -> cols);
    for(int i=0; i<this -> rows , i++)
    { for (int j=0; j< this->cols ; j++)
      {  temp.matrix [i][j] = 0;
         for(int k=0; k < this -> cols; k++)
         {
             temp.matrix [i][j] += this -> matrix [i][k] *
                                   other.matrix [k][j];
         }
    }
} return temp;
```